

Maximiliano Solorza Madrid

Software Engineering Student – Backend Systems, Cloud & Machine Learning

Software engineering student with production experience building backend services, relational database systems and cloud-deployed applications. Sole engineer on a multichannel AI support bot running on Google Cloud Run, and on a full-stack POS and inventory system delivered to a real client. Completed a software engineering internship at a U.S. healthcare technology company. Ranked in the top 12% globally at the IEEE Programming Marathon (924/8,169 teams). **Based in Turin, Italy for an exchange semester – available on site, no relocation required.**

maximilianoo.adonis@gmail.com linkedin.com/in/maximilianosolorza/ github.com/maxxee1 +569 7961 3993

Experience

Backend & Cloud Engineer (Freelance) Jul 2026 – Present
Movistar Game Club – multichannel AI support bot (sole developer) Santiago, Chile

- Sole engineer on a production multichannel (WhatsApp + Instagram) support bot: owned architecture, data model and deployment.
- Built an async FastAPI service with three-tier FAQ retrieval (keyword, fuzzy matching, embeddings), removing a 30s cold start.
- Designed a Redis-backed conversation state machine with token-budgeted memory and weighted round-robin agent routing.
- Integrated Gemini, BigQuery, Google Sheets and SendGrid behind a single channel-agnostic dispatch layer.
- Shipped a React + TypeScript admin panel (JWT auth, ticket triage, CSAT surveys) on Cloud Run, backed by 300+ automated tests.

Software Engineer (Freelance) Mar 2026 – Jun 2026
Fast-Food POS & Inventory System (sole developer) Santiago, Chile

- Built a full-stack POS and inventory management system with Next.js 16, React 19 and TypeScript, replacing manual ledger tracking.
- Engineered a profitability engine and combo simulator on PostgreSQL 18 and Prisma 7, modelling real margins, fixed costs, VAT and card commissions.
- Implemented recipe-based stock deduction so each sale automatically discounts ingredients by weight from inventory.
- Designed a local-first deployment with Docker Compose and automated AES-256-GCM encrypted backups, removing recurring cloud costs.

Software Developer Intern Dec 2025 – Mar 2026
Abacus Rx, Inc. Miami, United States

- Developed a dynamic reporting tool automating data extraction and visualization for internal pharmacy management systems.
- Optimized SQL queries and resolved performance bottlenecks, improving responsiveness in data-intensive operations.
- Investigated and resolved software incidents, debugging backend logic and database interactions in production environments.
- Contributed to development and testing of a mobile application integrated with the company’s pharmacy software ecosystem.
- Produced technical documentation for internal tools, system workflows and debugging procedures to support the team.

Teaching Assistant in Databases Mar 2025 – Dec 2025
Diego Portales University Santiago, Chile

- Supported two courses covering normalization, transactions, indexing, triggers and stored procedures.
- Designed and led weekly PostgreSQL laboratory sessions, building datasets, SQL exercises and hands-on query challenges.
- Debugged complex SQL queries and guided students in relational design and query optimization during lab sessions.
- Evaluated assignments and exams across two cohorts, providing technical feedback to support student learning.

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, C++, Java, Cypher
Backend: FastAPI, Node.js, Express, Next.js, REST APIs, async programming, JWT authentication
Data: PostgreSQL, Redis, BigQuery, Neo4j, Prisma, Pandas, NumPy, query optimization
Cloud & DevOps: Google Cloud Run, Docker, Docker Compose, Git, CI workflows, AWS (basics)
AI/ML: Neural networks, embeddings, Bayesian networks, Hidden Markov Models, TensorFlow/Keras, pgmpy, hmmlearn
Other: Linux (Ubuntu, Kali), Wireshark, Power BI, Jupyter, Matplotlib

Education

Politecnico di Torino – Exchange Semester Turin, Italy
Semester abroad, Computer Engineering coursework Sep 2026 – Feb 2027

Diego Portales University Santiago, Chile
B.Sc. in Information Technology & Telecommunications Engineering Expected Jul 2028

Competitions

Capture The Flag Hackathon – SEK Jun 2026

- Ranked 4th out of 24 teams (3,660 points).
- Contributed 2,180 points individually, ranking 11th among 75 participants.
- Solved cybersecurity challenges involving web vulnerabilities, cryptography and network analysis.

Programming Marathon – IEEE

- Achieved 6th place nationally in Chile and top 12% globally out of 8,169 teams.
- Solved algorithmic and data structure challenges during a continuous 24-hour competition.

Oct 2025

Certifications

Cisco Networking Academy: Introduction to Cybersecurity | Cybersecurity Essentials | Networking Essentials

[Credential]

Languages

Spanish (Native) | English (B2, professional working proficiency) | Italian (A1 – in progress)